

Juneha Hwang

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Education

Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)

Daejeon, South Korea

- **Relevant Coursework:** Linear Algebra for Data Science, Programming Principles, Data Structures, Introduction to Algorithms, Introduction to Networks, Introduction to Software Engineering, Machine Learning, Graph Machine Learning and Mining, Systems Programming, Computer Organization, Programming Languages

Experience

Linq Alpha

Jan 2025–Feb 2026

Data Engineer

Seoul, South Korea

- Engineered an entity resolution pipeline that maps ambiguous financial entity names to stock identifiers using a multi-stage agentic architecture (cache → trigram DB search → vector DB retrieval → LLM-based decision), combining RAG and structured reasoning for reliable production use.
- Built a semantic document chunking system that transforms unstructured financial documents (earnings calls, broker research, investment notes) into structured, queryable metadata using a multi-batch LLM pipeline—including multi-layer JSON parsing and validation to handle non-deterministic model outputs at scale.
- Designed and deployed a FastAPI-based API layer with asynchronous processing (Celery/Redis) for long-running LLM and data jobs, with observability instrumentation to monitor agent reliability in production.
- Managed large-scale financial datasets using PostgreSQL and Alembic, providing the retrieval backend for downstream AI pipelines.

Projects

DEX-DEX Perpetual Futures Arbitrage Bot | Rust, Tokio, WebSockets

- Built a low-latency arbitrage bot in Rust that detects and executes spread opportunities between two perpetual futures exchanges across 96 asset pairs in real time.
- Designed a dual-channel architecture using lock-free crossbeam channels for the hot path and Tokio broadcast for cold-path recording, with a dedicated OS thread for the signal engine to avoid async runtime contention.
- Implemented a maker-taker execution strategy with simultaneous post-only orders on both exchanges, sub-second fill polling, and automatic taker IOC on the opposite leg, including retry logic and reduce-only cleanup for failed fills.
- Optimized for cache performance with 64-byte aligned price state structs and atomic flags for position tracking, eliminating mutex contention on the critical path.

Papers

Structuring the Unstructured: A Multi-Agent System for Extracting and Querying Financial KPIs and Guidance

- Chanyeol Choi, Alejandro Lopez-Lira, Yongjae Lee, Jihoon Kwon, Minjae Kim, **Juneha Hwang**, Minsoo Ha, Chaewoon Kim, Jaeseon Ha, Suyeol Yun, Jin Kim
- Published at **FinIR Workshop @ ACM SIGIR 2025**, Padua, Italy

Honors

Competitive Programming

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025 Finalist
- International Collegiate Programming Contest (ICPC) Seoul Regional 2025 Finalist, Taichung Regional 2025 Finalist (Silver Medal)
- Union of Clubs for Programming Contests (UCPC) 2024, 2025 Finalist

Other Honors

- Jane Street ETC 2024 1st Place
- Fall 2023 Dean's List (KAIST)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, Typescript, Ruby, SQL, HTML/CSS

Tech Stack: FastAPI, PostgreSQL, Celery, Redis, React, Scikit-Learn, Numpy, Pandas, Git

Languages: Korean (native), English (native), Japanese (conversational)